SFU

MSE 312 Mechatronics Design II

Electronics Drive Circuit and Control **Summer 2026**

Class/tutorial hours/location: Wednesdays (2:30-4:20 PM), Fridays (2:30PM - 4:20PM)

Labs: Wed/Thu/Fri: 4:30-7:20 PM, SRYC 4080

The following topics will be discussed during classes with possible minor changes: Overview of lecture material; Electrical safety; Electronics for DC motor control; PWM motor drive; heat sink calculations; DC motor characteristics; speed/position control design.

Electronics Experiments

In this project, each group will design and build the power interface circuitry for driving a DC motor. The same motor can be used for actuating the mechanical arm designed in 1st part of the course.

• Objectives

a) Design and build a power drive circuitry for DC motor control by using available power MOSFETs and/or Bipolar Junction Transistors, H-bridge Integrated Circuit, and other components such as resistors, capacitors, logic gates.
b) Practice electronic test procedures to evaluate circuit operation in the lab using regular measurement and test equipment such as breadboards, function generators, oscilloscopes, power supplies, and multi-meters.
c) Build a working power electronic driver on a pre-drilled copper prototyping board (proto-board) or the custom PCB.

In next stages, the circuit will be used along with an embedded controller (Arduino MCU) and rapid prototyping tools such as Matlab/Simulink Real-time Target environment to build and test a feedback control system.

• Requirements

a) The circuit should be able to handle the DC motor ratings such as current, voltage, power, temperature (datasheets are posted on canvas).
b) Appropriate protection should be included in the design.
c) The circuit is expected to work in the normal room temperature (20-25 degrees Celsius). Proper heat sink should be designed for the devices. The circuit should be soldered.

d) The circuit should be able to drive the motor back and forth, with different speeds depending on the PWM duty cycle applied to the H-bridge

NOTES IN DOING THE ELECTRONICS:

- Make your logic circuits on the breadboard first. Your circuit should be able to rotate the motor in both directions.
- Before connecting the power supplies, make sure that all the connections are correct. You may want to use the Beep Test of the multi-meters for this part. Turn everything OFF when doing conductivity tests.
- In preliminary experiments, use an oscilloscope to observe the generated PWM signals with fixed duty cycles. Changing the duty cycle should affect the voltage across the motor terminals.
- After testing the circuit on a breadboard and verifying its operation, use the proto-board to solder all components. Neatness of the soldered circuit is important for the demo.
- Please record your measurements: Obtain the graph of duty cycle vs. terminal voltage for CW and CCW directions of motor rotation.
- Demos will be held during official lab sessions.

Control Project

The students will design a feedback system to control position/speed of a DC motor by utilizing linear control system methods. Please select reasonable numerical values (e.g., based on motor datasheet) for the parameters required to perform design and simulation studies.

Objectives

- Apply concepts in control system design process such as transfer functions, root-locus, frequency response, transient analysis.

- Conduct simulations and analyze and interpret data to improve the design.

- Apply design and test procedures using computer analysis, design, and simulation tools.

- Each group is required to come up with a set of requirements based on the system they choose. Students are encouraged to come up with their own ideas in terms of what they want to incorporate into the control design and how to simulate/experiment its performance.

Requirements

Make reasonable assumptions about the requirements of the control system in terms of response time, bandwidth, and how the feedback system responds to parametric variations (e.g., changes in motor, arm, electronics drive parameters).

PROCEDURE

- Obtain the transfer functions between the input voltage to the motor $EE_{aa}(ss)$ and the output angular position $\Theta_{mm}(ss)$ and velocity $\Omega_{mm}(ss)$ of the motor shaft.
- Use the datasheet for the motor (e.g., GM8224S009 DC Gearmotor- see datasheet posted on canvas) to obtain the necessary parameter values in control design such as torque constant, gear ratio, motor inertia, etc.
- Use MATLAB/SIMULINK software tools in support of your controller design (e.g., using root-locus or frequency response method).
- Obtain the time domain response and evaluate performance of the system in the presence of nonlinearities such as power drive saturation (e.g., ±12V saturation limits), friction, nonlinearities, and parametric uncertainties. Provide justification for any discrepancies

Lab Demonstration and Reporting

Each group must demonstrate the performance of their system with the Teaching Assistants and submit a written report through canvas by the due date. The Control System Design Report containing the following sections:

i. Introduction.
ii. Detailed analysis and design of the electronic circuitry
iii. Heat sink design
iv. Simulation/experiment results specifying the performance of the electronic circuit and control system, how the design meets requirements, any problems encountered, how they were addressed or would be addressed if you had more time.
v. Test procedures
vi. Conclusions.
vii. Code and simulation files as an attachment.

MSE 312 Mechatronics Design II

Electronics Drive Circuit and Control Design **Summer 2026**

Summary

- The electronics and control report (one per group) should be prepared in electronic format and submitted via canvas.sfu.ca before the deadline.
- All students in each group are expected to equally contribute to the completion of the project/report and will receive the same mark. If this is not the case, the students in each group should indicate the percentage contribution of each member. The individual marks will then be calculated based on the percentages indicated in the report.
- Details about the electronics/control design will be discussed in class. Students are required to attend the class and take notes. Hand-written lecture notes will not be posted on canvas.
- Mark breakdown is provided in the following table

Evaluation Component	**Details**
Electrical/Electronics: **See Lab Demonstration and Reporting for details**	Lab performance (20%): Includes attendance and organization in the labs, cleanliness of the circuit built, and successful demonstration of its operation.
Control System: **See Lab Demonstration and Reporting for details**	Lab performance (20%): Includes attendance and organization in the labs, cleanliness of the system built, and successful demonstration of control system operation.
Final Report	The intention of the final report for this part is thorough documentation of your work so far in regard to the motor drive system. Include comments regarding choices made, difficulties experienced and engineering design calculations. (60%). Rubric will appear on the Canvas Assignment.

Contact information:

- Teaching Assistants: Please see contact info on canvas

- Instructor (Control): Dr. M. Moallem (email: mmoallem@sfu.ca).
  - Office hours: Please make an appointment through email. We can meet in person or through zoom at https://sfu.zoom.us/j/9451219625